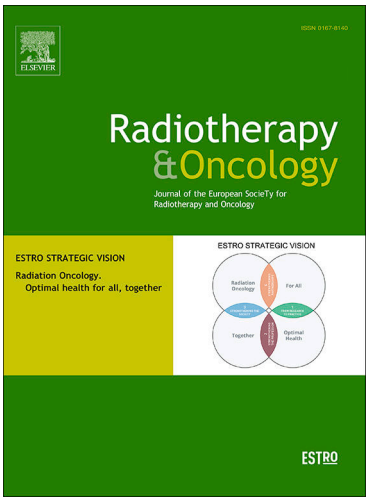
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Original Article

Multicenter deep Learning–Based automatic delineation of CTV and PTV in uterine malignancy CT imaging

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[Article Full Title]Multicenter Deep Learning–Based Automatic Delineation of CTV and PTV in Uterine Malignancy CT Imaging

[Short Running Title] Automated Segmentation Performance of nnU-Net Across Two-Center Uterine Malignancy Cohorts

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[Data Availability Statement for this Work] The code and model in this work will be available at https://github.com/TJ-MedRT/UMTVSeg. The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled access data storage at the Shanghai Tongji Hospital and Changzhou Fourth People's Hospital.

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Multicenter Deep Learning‐Based Automatic Delineation of CTV and PTV in Uterine Malignancy CT Imaging

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Background and Purpose:

Accurate delineation of the clinical target volume (CTV) and planning target volume (PTV) is essential for effective radiotherapy in uterine malignancies. Manual contouring is laborious, time-consuming, and subjective, and current automatic methods often focus on a single cancer type with limited external validation. To address this, we developed a deep-learning model capable of accurately delineating both CTV and PTV across multiple uterine malignancies using CT imaging.

Materials and Methods:

We retrospectively collected 602 contrast-enhanced CT scans, comprising 302 cases (cervical and endometrial cancers) from our institution and an additional 300 cervical cancer scans from external centers. Expert radiation oncologists manually delineated the CTV and PTV on each image. Among the 302 internal cancer cases, 177 cervical cancer cases were used for model training with five-fold cross-validation. Additionally, 41 cervical cancer cases were reserved as an internal testing cohort, while 84 endometrial cancer cases constituted the first external testing cohort to assess the model's generalizability across cancer types. The remaining 300 cervical cancer scans from external centers formed a second external testing cohort to assess model robustness across institutions. We evaluated three segmentation architectures—2D, full-resolution 3D, and cascaded 3D networks—and measured their performance using three standard metrics: Dice Similarity Coefficient (DSC), 95% Hausdorff Distance (HD95), and Average Surface Distance (ASD).

Results:

The model-generated segmentations demonstrated strong concordance with the expert contours. In the internal testing cohort with the same cancer type, performance metrics (DSC, HD95, ASD) were consistently high. Similarly, the external testing cohort with different cancer types showed robust performance, indicating effective generalizability. On the internal testing cohort, the model demonstrated strong performance, achieving mean DSCs of 83.42% for PTV and 81.23% for CTV, with low spatial errors (PTV: ASD 2.01 mm, HD95 5.71 mm; CTV: ASD 1.35 mm, HD95 4.75 mm). In the endometrial cancer cohort, PTV segmentation achieved a DSC of 82.88%, while CTV segmentation yielded an HD95 of 5.85 mm and an ASD of 1.34 mm. Additionally, clinical evaluation revealed that approximately 90 % of the model-generated contours required no or only

minor revision.
Conclusions:
We present a multicenter-validated deep-learning based framework for automatic CTV and PTV delineation across diverse uterine malignancies on CT. Our model offers a scalable, generalized solution with the potential to reduce the workload in radiation oncology, improve consistency, and streamline clinical workflows.

keywords: Uterine malignancy radiotherapy; Clinical target volume auto-delineation; nnU-Net; Multi-institutional validation

Introduction

Cervical cancer is one of the leading causes of cancer-related mortality in women worldwide, posing a significant challenge to global public health[1]. For stage IIB and more advanced cervical cancer, radiation therapy is a cornerstone of treatment[2]. Intensity-modulated radiation therapy (IMRT) is the most widely used radiation technique for cervical cancer, as it provides high-precision treatment doses to the tumor and reduces the dose to the organs at risk [3]. The principal aim of external beam radiotherapy planning is to maximize tumor dose while minimizing exposure to adjacent healthy tissues and organs. Accurate planning is essential: precise delineation of the clinical target volume (CTV) ensures adequate tumor coverage and spares surrounding tissues, thereby optimizing both treatment efficacy and patient safety[4]. Endometrial cancer, another common uterine malignancy, also relies on radiation therapy in its management, particularly in patients with high-risk or advanced-stage disease[5]. Cervical cancer is typically squamous cell carcinoma with direct local invasion, whereas endometrial cancer is usually adenocarcinoma and spreads mainly via lymphatic and peritoneal routes. Target volume delineation is generally similar, but the inclusion of presacral lymph nodes is standard for cervical cancer and endometrial cancer with cervical stromal invasion, and remains debated in other low-risk endometrial cancers.

According to consensus guidelines for radical treatment, CTV delineation should encompass the gross cervical tumor, entire cervix and uterus, parametrial tissue, upper vagina, ovaries, and relevant pelvic lymph node regions (common, external, internal iliac, obturator, and presacral nodes)[6]. However, manual CTV contouring by radiation oncologists is time-consuming and prone to inter- and intra-observer variability, which can lead to suboptimal treatment plans[7]. Moreover, the accuracy of CTV delineation relies heavily on the clinician's expertise and subjective judgment, leading to considerable inter-observer variability and potential inconsistencies in treatment planning. Similarly, delineation of the planning target volume (PTV) is essential to account for uncertainties such as organ motion and patient positioning. The PTV is typically generated by adding a margin around the CTV to ensure adequate coverage of the target area during treatment. Inaccurate or inconsistent PTV contouring can further compromise treatment outcomes, highlighting the need for automated, consistent, and accurate contouring methods to improve the quality and efficiency of radiation therapy planning.

Automated CTV delineation significantly accelerates the contouring process and enhances consistency. Numerous studies have explored automatic segmentation of anatomical structures from computed tomography (CT) and magnetic resonance imaging (MRI) scans[8]. Benefitting from the advantages of feature learning, deep learning (DL)-based segmentation methods have provided a promising solution to solve the problems of manual delineation and achieved end-to-end automatic

delineation[9-11]. Although these methods offer notable advantages in reducing manual annotation efforts and improving segmentation accuracy, these existing DL-based automatic delineation models have been developed based on a single type of cancer disease with strict inclusion and exclusion criteria, while it is unclear whether the segmentation method used on these datasets would produce the same results with more heterogeneous data. These task-specific designs of DL models still face challenges such as high memory consumption, risk of overfitting, and limited clinical applicability[12].

As an emerging universal solution, nnU-Net[13] leverages a self-configuring deep learning framework acclaimed for its automated parameter tuning and robust generalizability across diverse biomedical image segmentation tasks without task-specific design. We implemented this framework for CTV and PTV delineation in cervical cancer CT scans. Recognizing its potential, our goal was to develop a generalizable deep learning model capable of accurately segmenting target volumes in both cervical and endometrial cancer CT images. We compiled a dataset of 302 contrast-enhanced CT scans (including cervical and endometrial cancer cases from our institution) and 300 additional cervical cancer scans from a single external center, all manually annotated by experienced radiation oncologists. Using the nnU-Net framework, we benchmarked three U-Net variants, including 2D, full-resolution 3D, and 3D cascaded, to identify the optimal segmentation model. Model performance was evaluated using quantitative geometric metrics and subjective expert review for clinical feasibility. To examine generalizability, we further tested the model on endometrial cancer cases, confirming its applicability across different uterine malignancies.

Materials and Methods

Data Collection/Data

This retrospective study was approved by the ethics committees of Shanghai Tongji Hospital and Changzhou Fourth People's Hospital, and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all patients. We collected CT scans from a total of 602 patients: 518 cervical cancer cases (including both postoperative and definitive radiotherapy patients, from November 2014 to December 2023) and 84 endometrial cancer cases. CT scans were performed using a Philips MX16 scanner (Philips Healthcare, Best, the Netherlands) and a Toshiba Aquilion scanner (Canon Medical Systems, Otawara, Japan). Images were reconstructed with a matrix size of 512 × 512 pixels and a slice thickness of 5 mm. Patients were scanned in the supine position following bladder preparation, using both non-contrast and contrast-enhanced CT sequences to ensure accurate target delineation.

TNM staging for surgical patients was performed according to the latest AJCC (Version 9) guidelines, utilizing postoperative pathological findings for surgical cases and clinical staging for non-surgical patients [14]. In some cases, receiving neoadjuvant chemotherapy, downstaging of the T category was observed postoperatively. Of the tumors classified as "other" histologies, there were 2 neuroendocrine carcinomas, 3 sarcomas, 1 leiomyosarcoma, and 1 adenoid basal carcinoma. We assembled a comprehensive dataset comprising 175 cervical cancer CT scans for training, 41 cervical scans for internal testing, 84 endometrial cancer scans as an external testing cohort for disease generalizability, and 300 cervical cancer scans from external centers to evaluate institutional robustness. All CT images were acquired prior to radiotherapy to guide target delineation and treatment planning. In the external cervical cancer cohort, clinical stage distribution was as follows: stage I (n = 80), stage II (n = 80), stage III (n = 112), stage IV (n = 26), and unknown stage (n = 2).

Manual segmentation

The CTV and PTV were manually delineated by trained radiation oncologists as part of the standard clinical workflow prior to radiotherapy, serving as the ground truth for segmentation. Initial annotations were performed by junior oncologists and subsequently reviewed by senior oncologists, with the CTV defined based on consensus guidelines[15,16]. For postoperative radiotherapy, the CTV includes the tumor bed and pelvic lymphatic drainage areas, with possible extension to para-aortic nodes in high-risk cases. For definitive radiotherapy, the CTV covers the primary tumor region (uterus, cervix, parametria, and invaded structures) and lymphatic drainage areas, with consideration for para-aortic nodes if necessary. The PTV was generated by uniformly expanding the CTV by 0.5–1.0 cm to account for organ motion and setup variability, aligned with guidelines suggesting 7 mm for nodal margins and 1.5–2 cm for primary tumor volumes depending on image guidance use[17]. To ensure delineation quality, all contours were reviewed and modified if necessary by a committee of three experienced radiation oncologists, each with over 10 years of expertise in pelvic tumor radiotherapy, in accordance with the standard clinical practice of our institution.

Experimental Setting

Preprocessing

Prior to training, the dataset undergoes a series of preprocessing operations. The images are normalized using a CT Normalization scheme without the application of a mask for normalization. The median image size in voxels is maintained at 69x512x512 with a spacing of 5.0x0.9765x0.9765. The dataset is also subjected to augmentation techniques to enhance the robustness of the model. These include random rotations, scaling, and the introduction of Gaussian noise and blur. A multiplicative brightness transform and contrast augmentation are applied to simulate variations in image acquisition.

Network Architecture and Implementation

We built our model on nnU-Net, leveraging its self-configuring capabilities to automatically adjust network depth, batch size, learning rate, and input patch size based on the dataset's characteristics—thereby optimizing memory usage and contextual learning in 3D cervical cancer CT volumes . The 2D network employs a planar U-Net architecture to process individual slices of volumetric medical images, featuring a base number of features of 32 and a maximum of 512, with each encoder and decoder stage consisting of two consecutive 3×3 convolutional layers followed by ReLU activation functions and downsampling achieved through eight max-pooling layers with kernel sizes progressively increasing from 1×1 to 2×2, leveraging skip connections to preserve spatial details lost during downsampling and optimizing for datasets with high in-plane resolution and limited inter-slice correlation. The 3D low-resolution network, serving as the first stage of the cascade architecture, reduces computational burden by resampling input spacing to 5.0×1.48×1.48 mm^3, using a patch size of 40×224×224 voxels, 1×3×3 kernels in the first encoder stage to capture axial features and 3×3×3 kernels in subsequent stages, and six asymmetric max-pooling layers (e.g., 1×2×2, 2×2×2) to handle anisotropic spacing, generating coarse segmentation masks that provide global anatomical context for the second stage. The 3D cascade full-resolution network, inheriting the structure of the 3D full-resolution model and incorporating the coarse segmentation from the low-resolution stage, concatenates the original image data with the coarse masks in the decoder path, refining details at the original 5.0×0.98×0.98 mm^3 spacing using a patch size of 32×256×224

voxels, and balancing computational efficiency with segmentation accuracy through a two-stage cascade strategy to achieve precise delineation of small structures and fine boundaries.

The experimental setting

Our dataset included 175 cervical cancer cases for training–validation and testing cohorts of 425 scans (41 internal cervical, 84 endometrial, 300 external cervical), enabling robust generalizability assessment. The overall workflow is illustrated in Figure 1. The overall technical pipeline of this study is illustrated in Figure 2. Preprocessing involved clipping CT intensities to the 0.5–99.5th percentile of foreground voxels (to retain quantitative HU consistency in CT) and applying dataset-wide z-score normalization to suppress outliers . Each 3D volume was cropped to its non-zero bounding box and resampled via cubic interpolation to standardize voxel spacing while preserving spatial context . nnU-Net then automatically determined the optimal patch size, batch size, network depth, and learning rate based on the dataset "fingerprint" and available GPU memory—achieving an ideal balance between computational efficiency and spatial coverage . During training, we incorporated real-time augmentations (elastic deformations, gamma adjustments, and random rotations) to enhance model robustness. Segmentation outputs, generated slice-wise (2D/3D), were assembled into coherent 3D masks and refined via post-processing (e.g., retaining the largest connected component) to ensure anatomical continuity.

Evaluation metrics

The overall assessment of the automatic delineation performance was conducted using the Dice Similarity Coefficient (DSC) [18], which quantifies the spatial overlap between the automatic and manual contours. To provide a comprehensive evaluation, two additional representative boundary-based metrics were employed to assess the accuracy of the automatic segmentation, including the 95% Hausdorff Distance (95HD)[19] and Average Surface Distance (ASD). The 95HD measures the 95th percentile of all surface-to-surface distances between the predicted and ground truth contours, providing a robust estimate of the worst-case boundary deviation while reducing sensitivity to outliers. The ASD calculates the mean shortest distance between the surfaces of the predicted and reference segmentations, reflecting the overall geometric similarity of the contours.

Testing of the AI-assisted system in clinical setting

We evaluated the clinical relevance of our AI-assisted segmentation by comparing radiotherapy plans based on both automatic and manual contours, using a blinded review by a committee of three senior radiation oncologists (each with >15 years of experience and >500 cervical cancer cases managed). Each automatic contour was graded on a four-tier scale—No Revision (clinically ready), Minor Revision (minor edits with no clinical impact), Major Revision (significant correction required before planning), or Rejection (unacceptable—must be redrawn)[20].

Statistical analysis

Statistical analyses were performed using SPSS 27.0 (IBM, Chicago, USA). Baseline patient characteristics were compared between the training-validation and internal testing cohorts. The chi-square test was utilized for categorical variables, such as FIGO stage, T stage, and pathological type, while the independent samples t-test was applied to assess differences in the continuous variable of age.

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Results

The training-validation set and the testing cohort were comparable across most baseline characteristics, with significant differences observed only in FIGO/T stage and treatment intent. Detailed statistical results are presented in Table 1.

Figure 3 presents an axial CT slice from a FIGO 2018 stage IIB cervical cancer patient without lymph node involvement, showing overlaid CTV and PTV. The tumor margins are notably indistinct, and the CTV often merges with adjacent anatomical structures—such as the bladder, pelvic musculature, and nearby vasculature—making both accurate delineation and precise dose delivery particularly challenging.

We evaluated our model using five-fold cross-validation on 177 cervical cancer CT scans. Three nnU-Net architectures were compared: 3D full-resolution, 2D slice-level, and 3D cascaded U-Nets. As detailed in Table 2, the performance metrics were as follows. For CTV segmentation: the average DSC scores were 77.96, 77.59, and 77.83 for 3D full-resolution, 2D, and 3D cascade models, respectively; the corresponding 95HD were 9.17 mm, 10.81 mm, and 9.29 mm. For PTV segmentation: DSC scores were 80.60, 80.37, and 80.52; 95HD values were 10.08 mm, 10.85 mm, and 10.15 mm, respectively. CTV delineation proved challenging due to its complex boundaries, often merging with nearby anatomical structures. Notably, the 3D full-resolution model demonstrated superior robustness to high-contrast edge variations, resulting in more consistent and accurate segmentation relative to the other architectures.

The inference efficiency of the models was evaluated. The 2D slice-level U-Net required approximately 11 seconds per case, whereas the 3D full-resolution U-Net and the 3D cascaded U-Net took approximately 56 and 90 seconds per case, respectively. Although the 3D full-resolution model demands more computation time compared with 2D methods, it delivered the best overall segmentation performance. The 2D slice-level approach was faster but slightly less accurate, while the 3D cascaded model—designed to reduce memory usage—was the slowest due to its two-stage inference pipeline. Considering both accuracy and speed, the 3D full-resolution U-Net was selected as the final model for independent external testing.

To assess the model's generalizability, its segmentation performance was validated on both internal and external testing cohorts. As detailed in Table 3, the external validation included a multi-center cohort of 84 endometrial cancer cases and 300 cervical cancer CT scans. PTV segmentation showed relatively consistent results across cohorts. On the internal testing cohort, the model achieved an average DSC of 83.42%, an ASD of 2.01 mm, and a 95HD of 5.71 mm. Similar performance was observed in the endometrial cancer cohort (DSC: 82.88%, ASD: 1.86 mm, 95HD: 6.92 mm). However, in an additional multi-center cervical cancer CT cohort (n = 300), the mean DSC declined to 77.76%, accompanied by large standard deviations in spatial metrics (ASD: 3.20 ± 2.85 mm; 95HD: 21.88 ± 20.16 mm), indicating considerable boundary variability despite moderate volumetric agreement.

CTV segmentation demonstrated robust performance in both the internal testing and endometrial cancer cohorts, with mean DSCs of 81.23% and 79.68%, respectively. While the segmentation

accuracy declined in the multi-center cervical cancer cohorts (mean DSC: 61.48%, ASD: 7.84 mm, HD95: 28.46 mm), this variation likely reflects the increased anatomical and imaging heterogeneity across centers. These findings suggest that while the model demonstrates promising generalizability, further refinement is warranted to address the observed inter-institutional variability. The qualitative performance of nnU-Net segmentation is visually demonstrated in Figures 4 and 5 for the internal testing cohort and the endometrial cancer external testing cohort, respectively. Figure 6 illustrates a transverse slice comparing the auto-delineated CTV with the manually contoured CTV, demonstrating their close spatial alignment.

The predicted CTV contours for 84 endometrial cancer and 41 cervical cancer cases in the testing cohorts were subjectively evaluated by a panel of radiation oncologists. Approximately 90% of the nnU-Net-generated contours were rated as requiring "No Revision" or only "Minor Revision." A summary of the evaluation results is presented in Table 4. Post-hoc stratification, detailed in Table 5, revealed reduced segmentation performance for different disease types and a trade-off between volumetric overlap and surface accuracy depending on the treatment intent.

Discussion

CTV delineation is critical for radiotherapy planning but remains time-consuming. Although many convolutional neural network (CNN) models target organ segmentation, few have addressed automatic CTV delineation in uterine malignancies. To bridge this gap, we collected standardized CT scans from 302 in-house cervical and endometrial cancer patients and 300 external cervical cancer cases. We developed a 3D full-resolution nnU-Net for automatic CTV segmentation in cervical cancer and assessed its performance on endometrial cancer data as well.

The 3D full-resolution nnU-Net outperformed the 3D cascade and 2D UNet models in cervical cancer CTV delineation, achieving the highest mean DSC (80.60%) and the lowest ASD (3.20 mm). It also demonstrated robust performance across both internal and external testing cohorts. Oncologists' evaluations confirmed its superior clinical acceptability with minimal need for manual revision.

CTV delineation is more challenging than normal tissue segmentation due to its indistinct boundaries, which rely on anatomical context rather than clear visual cues. Unlike organs at risk (OARs), CTV requires higher-level semantic feature extraction. The 3D full-resolution nnU-Net automatically adapts to dataset characteristics and leverages high-resolution processing and multi-scale contextual information to accurately segment complex anatomical regions, even in areas with ambiguous or overlapping boundaries[13, 21]. While many models are trained predominantly on postoperative datasets—where CTVs are relatively uniform and standardized—curative-intent radiotherapy requires modeling a broader spectrum of CTV shapes, extents, and guidelines. Recognizing this, our training and testing cohorts intentionally include both postoperative and definitive-intent cases, ensuring that the model learns from and is validated across the entire clinical spectrum. Consequently, our dataset is robust and representative enough to enable reliable model performance in diverse real-world clinical scenarios.

Radiation oncologists consider subjective evaluation more effective than quantitative metrics for CTV delineation because it accounts for clinically critical and high-risk anatomical regions that may not be adequately captured by uniform pixel-based assessments [22]. Although delineation guidelines are continuously updated, the complexity of CTV and institutional variations has

prevented the establishment of a standardized consensus, resulting in variability across centers. Therefore, multicenter collaborative studies are essential to develop comprehensive and uniform standards for accurately evaluating CTV delineation [23].

Automatic CTV delineation significantly reduces contouring time in radiotherapy planning. Manual segmentation for cervical cancer typically takes 60–90 minutes. In contrast, the nnU-Net model generates initial contours within seconds. The process requires only approximately 10 minutes for manual adjustment and 5 minutes for data transfer, reducing the total time to about 15 minutes per case [13,21]. The AI-assisted workflow consists of three streamlined steps: image retrieval from Picture Archiving and Communication System (PACS), automated delineation using a two-stage nnU-Net pipeline to refine ROI boundaries and minimize observer variability, and final manual review within the treatment planning system [24]. This approach improves not only efficiency but also consistency and accuracy, particularly in anatomically complex regions, thereby enhancing the overall quality of radiotherapy planning [13,24-25].

To our knowledge, this is among the first studies to validate a unified deep learning model for simultaneous CTV and PTV segmentation across multiple uterine malignancies and clinical centers. The 3D full-resolution nnU-Net demonstrated accurate and clinically acceptable performance in both cervical and endometrial cancers, supporting its potential to enhance workflow efficiency and reduce inter-observer variability. However, performance variation across institutions suggests that generalizability remains a challenge, likely due to real-world clinical heterogeneity. Several factors likely contributed to this heterogeneity. First, the patient cohort was mixed, including both postoperative and definitive radiotherapy cases that required different CTV definitions. Second, even with standardized RTOG-based protocols, subtle inter-observer variability in delineation was evident, particularly in challenging regions like the parametria and vaginal cuff. Finally, variations in imaging protocols, including scanner types and contrast timing, could have altered soft tissue visualization. Collectively, these factors contribute to the observed domain gap between internal and external performance. Although the dataset includes multicenter cases, further prospective validation across broader populations is warranted.

Future directions include improving model generalizability by incorporating external CT datasets [26,27] through semi- or self-supervised learning frameworks [28]. Additionally, integrating foundation models such as the segment anything model (SAM) may enhance segmentation robustness and adaptability across various tumor types and imaging modalities [29,30]. In parallel with these research efforts, a clear clinical integration pathway has been proposed, centered on an AI-assisted three-step workflow (image retrieval from PACS, AI-based delineation, and manual review) designed to reduce physician workload and improve consistency. This workflow has been demonstrated to significantly reduce manual contouring time from 60–90 minutes to approximately 15 minutes, with approximately 90% of its generated contours requiring no or only minor revision, demonstrating high clinical utility. Subsequently, paving the way for clinical adoption requires large-scale prospective validation to assess the model's performance and reliability across varied patient cohorts and settings. These findings are anticipated to be of broad interest to experts in radiation oncology, medical imaging, and clinical artificial intelligence, especially those focused on automating and standardizing radiotherapy planning in heterogeneous clinical settings. Further investigation into domain adaptation techniques will be essential for improving performance on out-of-distribution data.

Conclusions
We developed a 3D full-resolution nnU-Net for automatic CTV segmentation in cervical and endometrial cancers. Validated on multi-center data, it outperformed 2D and cascade models, delivering accurate, clinically acceptable results and reducing contouring time from over an hour to about 15 minutes. This highlights its potential to improve efficiency and consistency in radiotherapy planning for gynecologic cancers.

Abbreviations
CTV: Clinical Target Volume; PTV: Planning Target Volume; DSC: Dice Similarity Coefficient; HD95: 95% Hausdorff Distance; ASD: Average Surface Distance; CT: Computed Tomography; MRI: Magnetic Resonance Imaging; ROIs: Regions of Interest; CNN: Convolutional Neural Network; OARs: Organs at Risk; PACS: Picture Archiving and Communication System; SAM: Segment Anything Model

Declarations
Conflicting Interests
No.

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Authors' contributions
Bichun Xu led model development and manuscript preparation. Jun Liu and Mingming Fang performed data collection and processing. Hong Zhu, Xujing Lu and Judong Luo conducted the clinical validation. Yichi Zhang and Hongwei Zhang supported external data evaluation. Xujing Lu and Judong Luo supervised the study and revised the manuscript. All authors reviewed and approved the final version.

Data Availability
The code and model in this work will be available at https://github.com/TJ-MedRT/UMTVSeg. The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request. Data are located in controlled-access data storage at the Shanghai Tongji Hospital and Changzhou Fourth People's Hospital.

Ethics approval and consent to participate
Prior to enrollment and treatment, all patients signed an informed consent form and received thorough information from their oncologists regarding the potential toxicity and benefits of radiotherapy. This retrospective study was reviewed and approved by the Ethics Committees of Shanghai Tongji Hospital and Changzhou Fourth People's Hospital, and it was conducted according to the Declaration of Helsinki.

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Consent for publication
Not applicable.

Declaration of interest
The authors declare that they have no competing interests.
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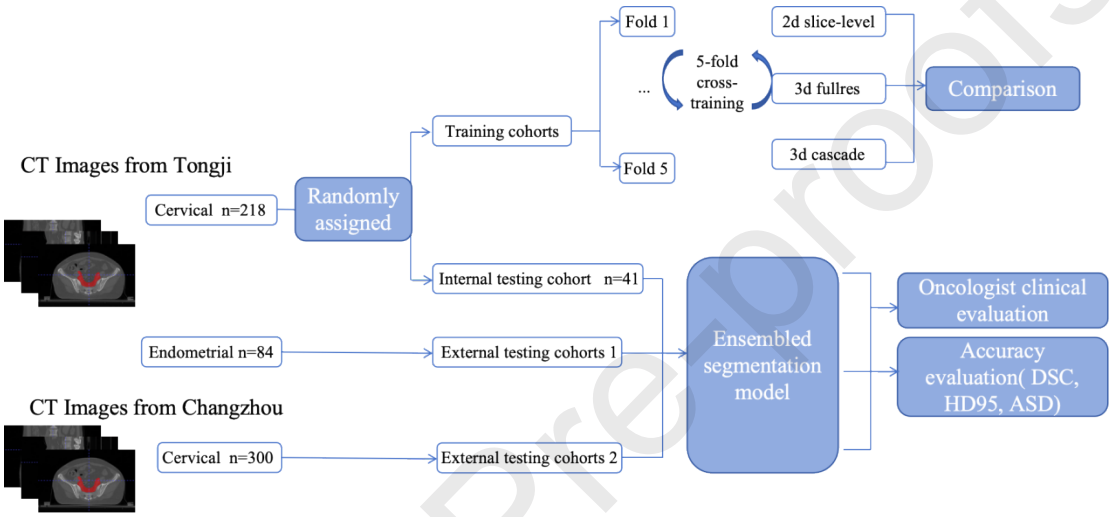


Figure 1. Network training and sketching verification flow chart.

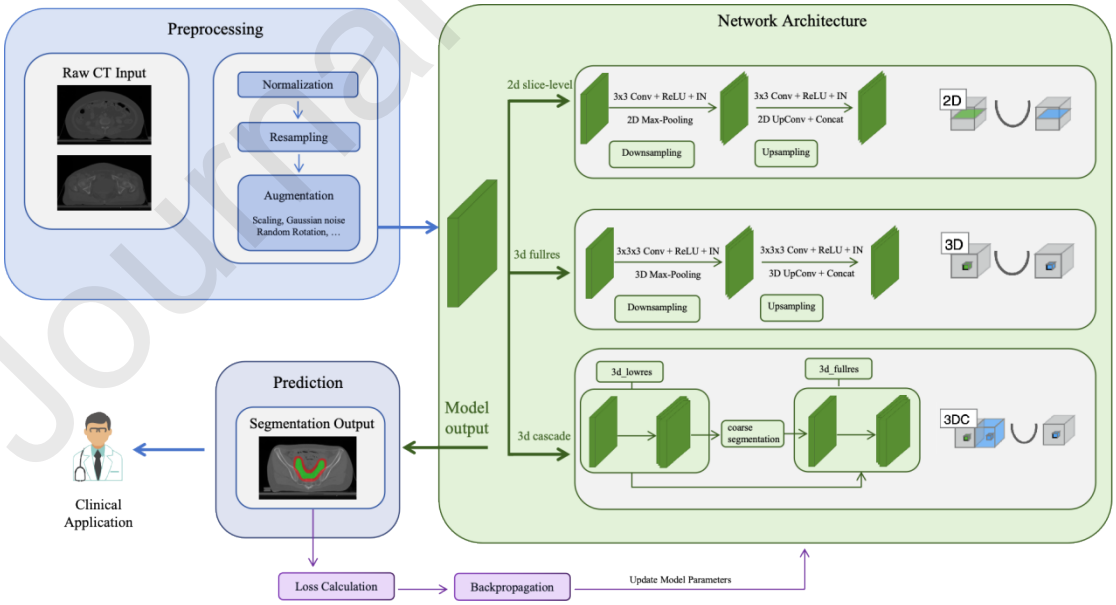


Figure 2. The architecture of proposed nnU-Net 3D fullres, 2D slice-level, and 3D cascade models.

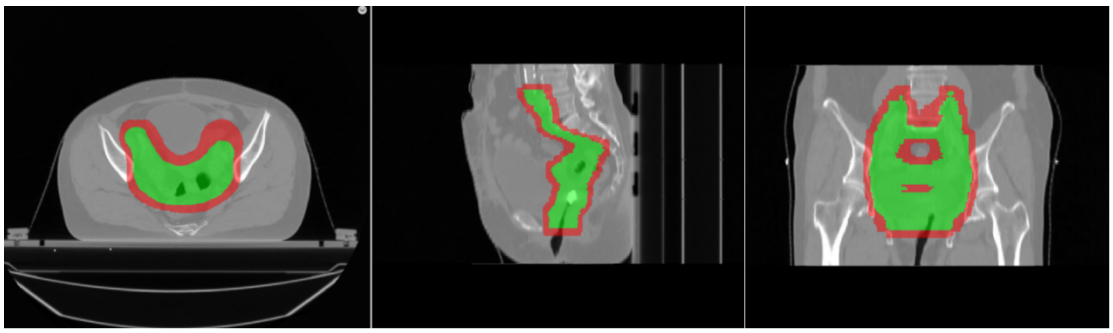


Figure 3. CT axial image of a FIGO 2018 stage IIb cervical cancer patient with delineated CTV & PTV.

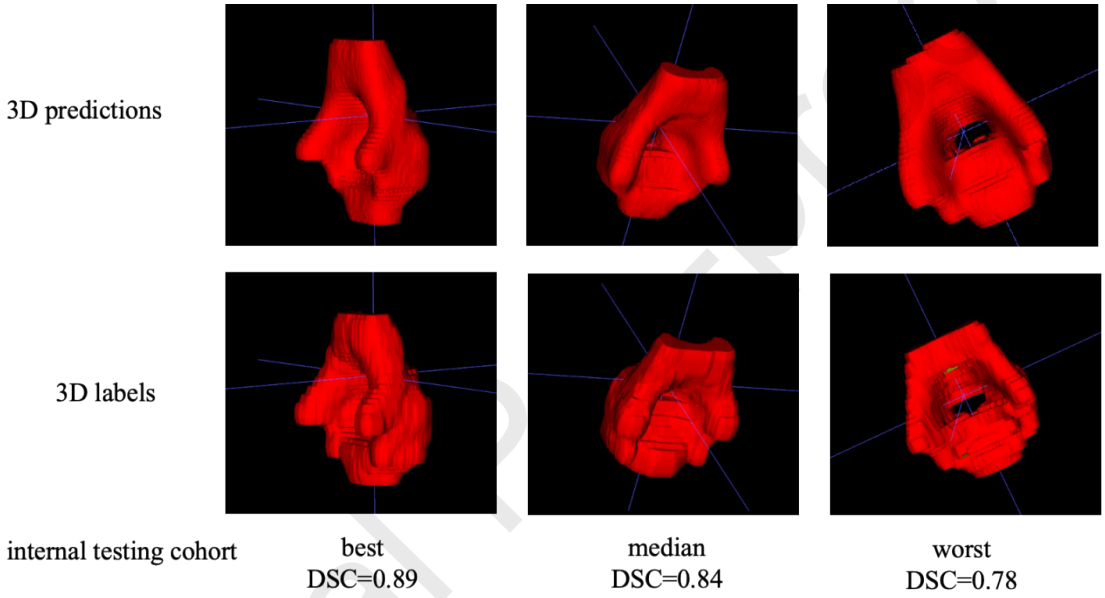


Figure 4. Qualitative assessment of nnU-Net segmentation on internal testing cohort. 3D comparison of nnU-Net predictions (top row) and expert-annotated ground truth labels (bottom row) for internal test cases. From left to right, the cases represent the best, median, and worst segmentation performance based on the DSC.

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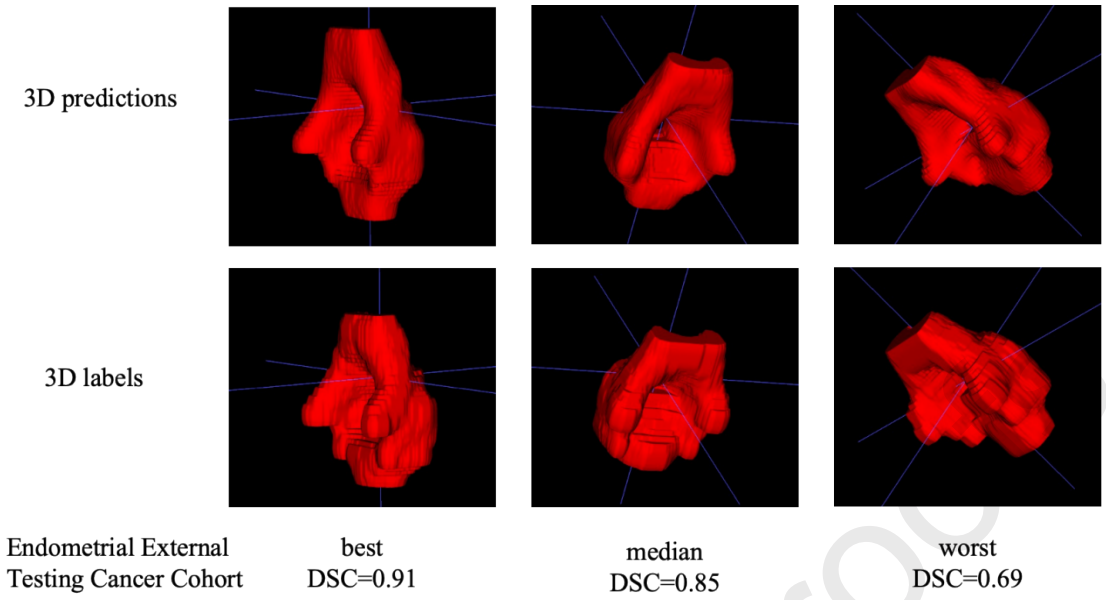


Figure 5. Qualitative assessment of nnU-Net segmentation on endometrial external testing cancer cohort. Representative 3D visualizations comparing nnU-Net predictions (top row) and expert-annotated ground truth labels (bottom row) for endometrial cancer cases. From left to right, examples represent the best, median, and worst segmentation results based on DSC in the external endometrial cancer test set.

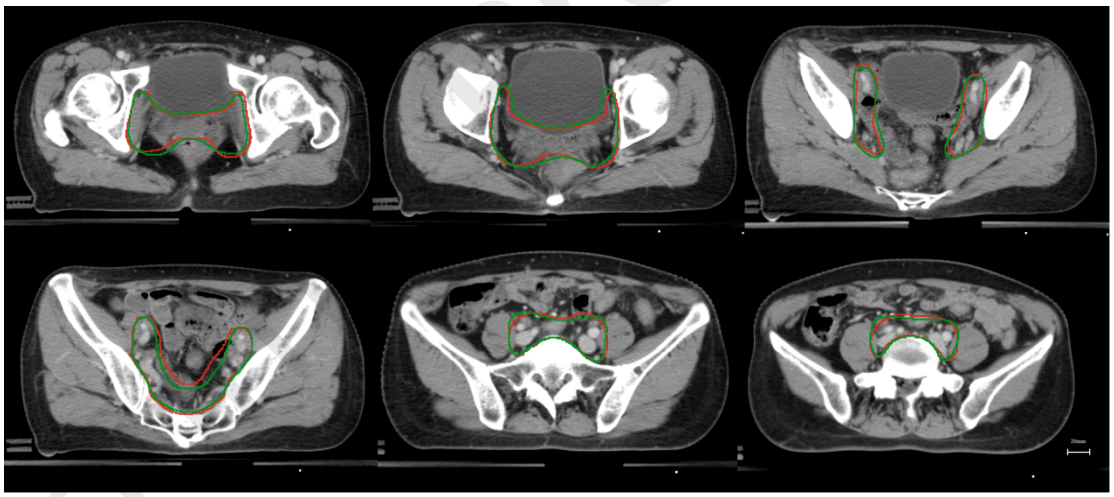


Figure 6. CTV delineation in a testing cohort cervical cancer patient. The green region represents the auto-delineated CTV, while the red line denotes the physician-delineated CTV.

Table 1. Baseline characteristics and parameters of the single-center training and testing cohorts

(cervical cancer dataset).

Characteristic		Training-validation cohort	Internal testing cohort	P value
n		175	41	
Age, median(range)		52（27-87）	57（38-75）	0.809
Pathological types	Squamous Cell Carcinoma	149	35	0.615
	Adenocarcinoma	11	3	
	Adenosquamous Carcinoma	8	3	
	Others	7	0	
Figo stage	I	68	9	0.004
	II	66	11	
	III	32	19	
	IV	9	2	
T	T1	91	12	0.000
	T2	67	13	
	T3	9	14	
	T4	8	2	
N	N0	122	29	0.785
	N+	53	12	
Purpose	postoperative	164	33	0.007
	definitive	11	8	

Table 2. nnU-Net models for one-step CTV & PTV segmentation (3D-fullres, 2D, 3D-cascade).

Method	Target	Metric	fold 0	fold 1	fold 2	fold 3	fold 4	Average
3D fullres	PTV	DSC	80.95±14.58	80.25±15.83	78.84±20.31	78.80±20.12	84.17±6.27	80.60
		ASD	2.64±4.70	2.49±3.79	7.52±22.80	3.22±7.07	1.87±1.24	3.55
		95HD	7.77±10.13	8.12±8.70	17.33±45.23	10.92±23.94	6.24±5.69	10.08
	CTV	DSC	76.29±16.83	78.60±15.13	77.57±19.81	77.20±20.19	80.16±7.61	77.96
		ASD	2.66±5.62	2.07±4.38	7.02±22.90	2.79±8.69	1.46±1.24	3.20
		95HD	7.98±10.81	6.36±8.13	15.98±45.38	9.63±23.40	5.88±6.29	9.17
2D slice-level	PTV	DSC	80.42±14.74	80.08±15.58	78.96±19.98	78.82±19.55	83.58±5.74	80.37
		ASD	4.88±16.15	2.69±4.15	7.56±22.78	3.34±7.06	1.90±0.74	4.07
		95HD	11.92±32.18	8.09±9.60	17.28±45.23	10.74±23.94	6.21±4.76	10.85
	CTV	DSC	75.94±17.01	78.29±15.20	77.46±19.55	76.67±20.34	79.60±7.41	77.59
		ASD	4.72±16.34	2.16±4.44	7.01±22.91	5.58±18.15	1.43±0.74	4.18
		95HD	12.13±32.63	6.34±8.21	16.02±45.39	13.94±38.89	5.64±5.33	10.81
3D cascade	PTV	DSC	80.62±14.55	79.91±15.89	78.79±20.24	79.16±18.69	84.10±6.17	80.52
		ASD	2.61±4.43	2.57±3.94	7.58±22.78	3.59±7.57	1.81±0.91	3.63
		95HD	8.57±14.78	8.29±9.03	17.44±45.22	10.30±21.96	6.14±4.81	10.15
	CTV	DSC	76.04±16.63	78.12±15.15	77.67±19.71	77.38±19.22	79.92±7.47	77.83
		ASD	2.68±5.56	2.14±4.52	7.08±22.89	3.23±9.11	1.37±0.85	3.30
		95HD	9.12±16.12	6.48±8.33	15.96±45.39	9.31±22.24	5.58±5.33	9.29

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Table 3.Segmentation performance metrics for PTV & CTV.

Dataset	Segmentation Type	Mean DSC (%)	ASD (mm)	HD95 (mm)
Internal Testing Cohort	PTV	83.42±3.65	2.01±0.54	5.71±1.85
Endometrial Cancer External Testing Cohort		82.88±7.77	1.86±0.86	6.92±6.76
Cervical Cancer External Testing Cohort		77.76±8.06	3.20±2.85	21.88±20.16
Internal Testing Cohort	CTV	81.23±4.40	1.35±0.46	4.75±1.57
Endometrial External Testing Cancer Cohort		79.68±9.29	1.34±0.67	5.85±6.60
Cervical Cancer External Testing Cohort		61.48±7.89	7.84±4.38	28.46±26.81

Table 4. Oncologists' subjective evaluation of the CTV contours.

Grade	External testing cohort n=84	Internal testing cohort n=41
No revision	23	12
Minor revision	53	25
Major revision	6	4
Rejection	2	0

Table 5. Segmentation performance metrics for PTV &CTV segmentation stratified by clinical subgroups.

Dataset		Segmentation Type	Mean DSC (%)	ASD (mm)	HD95 (mm)
Figo stage	I	PTV	75.30 ± 3.68	4.24 ± 3.93	18.25± 20.25
	II		77.09 ± 9.51	4.45 ± 4.15	21.02 ± 22.74
	III		77.04 ± 7.40	3.99 ± 3.50	19.73 ± 19.13
	IV		74.50 ± 9.22	5.83 ± 4.41	33.71 ± 24.63
Figo stage	I	CTV	61.86 ± 8.15	7.03± 4.36	22.73 ± 23.01
	II		65.06 ± 11.43	7.39 ± 4.82	25.91 ± 28.22
	III		66.34 ± 9.82	6.92 ± 4.33	25.14 ± 22.25
	IV		64.73 ± 10.40	8.67± 4.86	32.34 ± 20.87
Purpose	postoperative	PTV	75.36 ± 7.69	3.91± 3.65	17.56 ± 19.39
	definitive		78.37 ± 6.63	4.83 ± 4.19	25.06 ± 22.99
Purpose	postoperative	CTV	63.25 ± 11.19	6.67 ± 4.30	22.86 ± 24.67
	definitive		67.18 ± 7.06	7.93 ± 4.80	28.72 ± 23.57

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highlights（bullet points)

- Deep learning model segments CTV and PTV across uterine malignancies
- Multi-center validation demonstrates strong generalizability and robustness
- Model achieved high DSC and low surface errors on internal and external datasets
- Approximately 90% of contours required no or minor revision in clinical expert review

[Conflict of Interest Statement for All Authors] Conflict of Interest: None

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